ACIDS AND METALS 21

The general equation for the reaction of acids with metals is:

acid + reactive metal → salt + hydrogen

Note that not all metals react with acids. The most common acids used in a school laboratory are hydrochloric acid, nitric acid and sulfuric acid.

1 Name two metals that tend to not react with dilute acids.

2 Give the formula for each acid.

a hydrochloric acid **b** nitric acid **c** sulfuric acid

A **salt** is an ionic compound that is formed when an acid is wholly or partly replaced by a metal (or a metal radical). When a salt is formed, the reactive metal replaces the hydrogen in the acid. This hydrogen is given off as a gas.

3 Give the name of the salt formed by each acid.

a hydrochloric acid **b** nitric acid **c** sulfuric acid

4 Describe the test used to identify hydrogen gas.

Acids are polar covalent compounds that form ions when they dissolve in water.

$HCl\ (aq) + H_2O\ (l) \rightarrow H_3O^+\ (aq) + Cl^-\ (aq)$

$H_2SO_4\ (aq) + 2H_2O\ (l) \rightarrow 2H_3O^+\ (aq) + SO_4^{2-}\ (aq)$

$HNO_3\ (aq) + H_2O\ (l) \rightarrow H_3O^+\ (aq) + NO_3^-\ (aq)$

H_3O^+ is the hydrated hydrogen ion (sometimes called a hydronium ion): $H^+ + H_2O \rightarrow H_3O^+$. This ion is what gives a typical acid its acidic properties.

5 Name each ion.

a Cl^- **b** SO_4^{2-} **c** NO_3^-

This reaction shows ions in solution when magnesium reacts with hydrochloric acid:

$Mg\ (s) + 2H_3O^+\ (aq) + 2Cl^-\ (aq) \rightarrow Mg^{2+}\ (aq) + 2H_2O\ (l) + H_2\ (g) + 2Cl^-\ (aq)$

6 A *spectator ion* does not undergo any change in a reaction. Identify the spectator ion that occurs on both sides of the equation.

As with mathematical equations, terms that appear on both sides of the equation can be removed to give:

$Mg\ (s) + 2H_3O^+\ (aq) \rightarrow Mg^{2+}\ (aq) + 2H_2O\ (l) + H_2\ (g)$

or $Mg\ (s) + 2H^+\ (aq) \rightarrow Mg^{2+}\ (aq) + H_2\ (g)$

This is called a 'half-equation'.

7 When magnesium metal reacts, it changes from Mg to Mg^{2+}. Has it gained or lost electrons?

8 When a hydrogen ion reacts, it changes from H^+ to $\frac{1}{2}H_2$. Note that the convention is to show a hydrogen atom as half of a hydrogen (H_2) molecule. Has the hydrogen ion gained or lost an electron?

This reaction shows ions in solution when zinc reacts with sulfuric acid:

$Zn\ (s) + 2H_3O^+\ (aq) + SO_4^{2-}\ (aq) \rightarrow Zn^{2+}\ (aq) + 2H_2O\ (l) + H_2\ (g) + SO_4^{2-}\ (aq)$

Removing spectator ions gives:

$Zn\ (s) + 2H_3O^+\ (aq) \rightarrow Zn^{2+}\ (aq) + 2H_2O\ (l) + H_2\ (g)$

9 Identify the ion that occurs on both sides of the equation (i.e. the 'spectator ion').

10 When zinc metal reacts, it changes from Zn to Zn^{2+}. Has it gained or lost electrons?

11 Write ionic half-equations to show the reactions between each pair of substances.

a sulfuric acid and magnesium

b hydrochloric acid and zinc

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